# Adaptive Self-Supporting Prototype Learning for Remote Sensing Few-Shot Semantic Segmentation

Weihao Shen, *Graduate Student Member, IEEE*, Ailong Ma, *Senior Member, IEEE*, Junjue Wang, *Student Member, IEEE*, Zhuo Zheng, *Member, IEEE*, and Yanfei Zhong, *Senior Member, IEEE*

***Abstract*— The semantic segmentation of remote sensing images with few shots has important theoretical and application value. Most of the existing few-shot semantic segmentation frameworks are based on prototype learning methods, in which a single support prototype is designed to guide the query set for prediction. However, the visual differences between the support set and the query set make it difficult for a single support prototype, generated from the support set, to comprehensively encapsulate the semantic information of all the query images. This article introduces an adaptive self-supporting prototype learning network designed for few-shot segmentation (FSS), in order to tackle the challenges mentioned earlier. We propose adaptive hyperprototype representation (HPR), which consists of hyperprototype clustering (HPC) and guided prototype matching (GPM), to generate and assign multiple representative prototypes to compensate for the limitations of a single prototype in representing the semantic information of the query images. Specifically, HPC is a parameter-free and adaptive approach, which can extract more representative prototypes by aggregating similar feature vectors utilizing superpixel feature clustering. Meanwhile, GPM can select matched prototypes to provide more accurate guidance, allowing for uniformly aligned representation of multiple prototypes and complex image semantic information. We also introduce self-supporting matching (SSM) prototype learning, which can accurately guide the query set segmentation by acquiring query set prototypes. SSM generates initial pseudo labels for the query set based on the support set prototypes, and further guides the query set using the pseudo labels, along with the query prototypes generated by its own features, thus effectively avoiding visual differences between the support set and query set. The proposed adaptive self-supporting prototype learning network substantially improves the prototype quality and achieves a superior performance on object-level remote sensing datasets.**

***Index Terms*— Few-shot learning, few-shot segmentation (FSS), prototype learning, remote sensing, semantic segmentation.**

Manuscript received 20 December 2023; revised 14 April 2024, 16 May 2024, and 24 June 2024; accepted 19 July 2024. Date of publication 29 July 2024; date of current version 8 August 2024. This work was supported by the National Natural Science Foundation of China under Grant 42171336 and Grant U22A20567. *(Corresponding author: Ailong Ma.)*

Weihao Shen, Ailong Ma, Junjue Wang, and Yanfei Zhong are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: whshen@whu.edu.cn; maailong007@whu.edu.cn; kingdrone@whu.edu.cn; zhongyanfei@whu.edu.cn).

Zhuo Zheng is with the Department of Computer Science, Stanford University, Stanford, CA 94305 USA (e-mail: zhuozheng@cs.stanford.edu).

Digital Object Identifier 10.1109/TGRS.2024.3435086

## I. Introduction

THE analysis and understanding of remote sensing images can be approached from various levels and perspectives [1], [2], [3], [4], [5], [6].

Remote sensing image semantic segmentation is a crucial task in the field of remote sensing image processing. It aims to assign each pixel in the remote sensing image to its corresponding semantic category, enabling advanced understanding and analysis of remote sensing data. Over the past few years, the rapid advancements in deep learning technologies have yielded significant achievements in various computer vision tasks, including remote sensing image analysis. However, the traditional deep learning methods typically demand a substantial amount of annotated data for training, posing challenges in handling high-dimensional and large-scale remote sensing data. Acquiring a sufficient volume of labeled data for remote sensing images can be prohibitively expensive and time-consuming. If few-shot data are used directly to allow the traditional models to learn, this makes it difficult for the models to adequately learn the different classes of features. In the traditional methods, large-scale datasets are required to train the complex deep networks, which is often difficult to achieve in a few-shot context. Moreover, when faced with new scenes and novel categories, the conventional deep learning semantic segmentation models often fail to generalize effectively to the segmentation of these novel categories. These models are overly dependent on specific data distributions and sample sizes and are unable to adapt to new few-shot data.

In this context, Shaban et al. [7] were the first to propose few-shot segmentation (FSS) and establish a task setup and evaluation protocol. FSS has since been vigorously developed to address the problems of systematic sparse samples and novel classes. Specifically, FSS uses a few support samples to segment any of the novel classes. The current approaches use the prototype learning approach to match the prototypes obtained from the support set to the query set for prediction. The prototype learning techniques involve compressing masked object features within a support image into a single or few prototypical feature vectors. Subsequently, these techniques are employed to identify the corresponding pixel locations with similar features in the query image, facilitating the segmentation of the desired objects. However, land objects

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

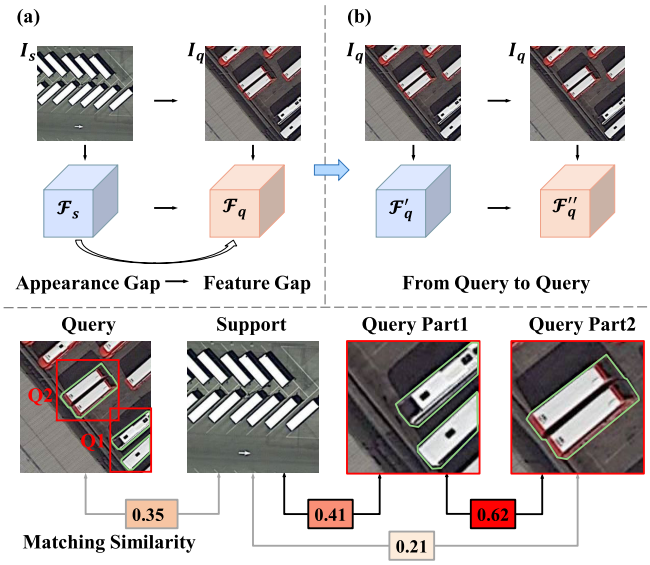



Fig. 1. (Top) Image illustrates the conventional approach, which is to guide the query set from the support set, where there is a feature gap between the support set and the query set, due to the appearance gap, so we consider guiding from the query set itself. (Bottom) Image illustrates self-supporting matching: objects from the same regions are more similar than objects from different regions. (a) Conventional support. (b) Query self-support.

are complex and diverse in remote sensing imagery, and more than a single or few prototypes are needed to express the semantic information of features comprehensively. The support images are also finite and fixed, while the query images are infinite and diverse. Regardless of the quality of the support set, the limited few-shot support quickly fails to cover the underlying appearance variations of the target classes in the query image. As shown in Fig. 1, there is a visual appearance gap between the images of the query set and the support set, resulting in the features generated by the model also having representational differences, making it impossible to decipher the query set well when guided by the support set. These are the two long-standing problems in few-shot learning caused by the inherent data scarcity and diversity. Researchers have followed the FSS setting to propose an FSS framework based on prototype learning and metric learning. This prototype learning approach performs metric segmentation of the queries by learning class prototypes [8]. Various prototype learning methods have also been proposed to obtain more robust prototype representations, such as masked average pooling-based [9] prototype learning and feature clustering-based [10] prototype learning. Although these methods have achieved good results, they still do not fundamentally solve the support and query appearance difference problem. The few-shot support also limits the ability to obtain more diverse and robust prototype representations.

In order to alleviate the above problems and to fill the research gap in the field, we propose adaptive hyperprototype representation (HPR) and self-supporting matching prototype learning for the few-shot framework of remote sensing images, which is referred to as SSMP. HPR alleviates the problem of a single prototype being unable to express the semantic information of complex and diverse remote sensing ground objects and specifically uses the idea of superpixel clustering on the extracted features to obtain the superpixel centroids. This guides the generation of multiple hyperprototypes to express the semantic information. The self-supporting matching prototype learning mitigates the problem of prototype differences caused by the difference in visual appearance between the support and query sets by generating query set pseudo prototypes for its guiding and matching, which is motivated by the classical Gestalt Law [11], i.e., pixels belonging to the same object are more similar than those belonging to different objects. Specifically, as shown in the bottom half of Fig. 1, some of the vehicles in the query set are more similar to the other vehicles than the query set is to the vehicles in the support set. Compared with the previous methods, SSMP can extract more robust prototypes, support prototypes, and self-support query prototypes. The proposed model has superior robustness and generalization, and it can better adapt to the diversity of remote sensing images and the complexity of ground objects. It also makes up for the difference in the appearance of the query sets and support sets and improves the segmentation performance of remote sensing images for agnostic classes in the query set. All these modules are highly conducive to improving the recall of segmented objects and suppressing irrelevant interference terms. The main contributions of this work are as follows.

### A. FSS Framework

To tackle the problem of remote sensing image segmentation in low-data regimes, we designed an adaptive self-supporting prototype learning FSS framework. The framework seamlessly fuses the HPR with the self-supporting matching prototype learning of the query sets, allowing the model to generate query hyperprototypes combining some of the diversity of the multiple prototypes and the similarity of its own guidance, thus generating more robust multiprototypes for the guided query set segmentation, to promote the performance of segmentation models in low-data scenes.

### B. Adaptive HPR

The HPR utilizes foreground features corresponding to the support or query image to learn multiple hyperprototypes for a more comprehensive representation of complex ground objects in remote sensing by dividing the image pixels into adjacent regions. The corresponding regional features are aggregated, so that each region produces multiple significant foreground support prototypes. This module can be arbitrarily embedded in the support set or query set of the framework for extracting foreground prototypes.

### C. Self-Supporting Matching Prototype Learning

To obtain prototypes of the query sets to avoid the differences between the support and query sets, and without using the ground truth (GT) of the query sets, we propose a pseudo-label self-supporting matching prototype learning approach. This predicts the query set by utilizing the support prototypes extracted in the previous stage and obtains preliminary query

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

set prediction results as pseudo labels for the query set prototype extraction. The query set extraction is also embedded in the previous module for HPR to obtain multiple robust foreground prototypes. The query set background prototypes are extracted by utilizing the adaptive self-supporting background prototype (ASBP) approach to perform multiprototype extraction, and finally obtain the query set foreground prototypes and background prototypes for query set prediction.

The proposed SSMP method was applied to large-scale high-resolution remote sensing imagery semantic segmentation datasets. Both comparative experiments and ablation experiments were conducted under the same experimental settings. The experimental results indicate that SSMP outperforms the state-of-the-art few-shot semantic segmentation methods.

## II. Related Work

### A. Semantic Segmentation

Semantic segmentation is an essential task in computer vision that aims to assign each pixel in an image to a predefined semantic category. In the past few years, many deep learning methods have made significant progress in the semantic segmentation task. With the development of deep learning, convolutional neural networks (CNNs) have become the dominant approach in semantic segmentation. Semantic segmentation methods based on CNNs usually include an encoder–decoder architecture, where the encoder is used to extract image features, and the decoder is used to generate the pixel-level semantic segmentation results. For example, fully convolutional networks (FCNs) [12], U-Net [13], the high-resolution network (HRNet), the pyramid scene parsing network (PSPNet) [14], and the DeepLab [15] family are commonly used CNN-based semantic segmentation methods [16], [17], [18] that have performed very well in the field of remote sensing segmentation. However, the performance of the general purpose semantic segmentation methods can be limited due to the need for a large number of labeled samples for training, particularly in few-shot scenarios characterized by data scarcity or specific scenarios.

### B. Few-Shot Learning

Few-shot learning is a challenging task in the field of machine learning [19], [20], [21], [22], [23], where the goal is to recognize and generalize from a limited number of training examples. Metric learning-based methods aim to learn a similarity metric that can effectively compare and classify unseen classes based on the few available samples. Prototypical networks introduced the concept of learning a class prototype to represent each class, enabling efficient distance-based classification in few-shot scenarios. Matching networks [24] employed an attention mechanism to weigh the importance of support samples during classification dynamically. Meta-learning or "learning to learn" methods have gained significant attention in few-shot learning. Model-agnostic meta-learning (MAML) [25] introduced a meta-learning framework to adapt a base model's parameters to new tasks with minimal training samples. In addition, the Reptile algorithm introduced a simpler optimization-based approach for fast adaptation to new tasks. In addition, generative models have been extended to few-shot learning, with methods leveraging generative models to synthesize additional training samples. Conditional variational autoencoders [26] and generative adversarial networks [27] have also been adapted for generating novel samples for few-shot learning, aiding in better generalization. There have been several related studies in remote sensing image processing; however, most of these methods predominantly concentrate on scene classification tasks.

### C. Few-Shot Segmentation

FSS is an emerging and promising research direction in computer vision, which primarily addresses dense prediction tasks using limited annotated data [28], [29]. The pioneering work by Shaban et al. [7] in 2017 played a pivotal role in defining the task settings, establishing standard benchmarks, and introducing evaluation criteria, thus paving the way for subsequent research in this domain. In recent years, the FSS field has witnessed significant progress, with several innovative approaches leveraging the advantages of few-shot learning, leading to remarkable performance improvements [30], [31] [32]. Wang et al. [33] introduced the prototype alignment network to address the FSS problem by aligning the query set and support set metrically, employing a metric learning perspective. They facilitated mining solely from the support set dataset through mapped prototype alignment regularization [33]. However, this method relies on a single prototype approach, generating only one foreground–background prototype for the support set. In contrast, Li et al. [34] proposed a more diversified approach for generating support set prototype representations using a superpixel-based method. This technique extracts highly representative prototypes by employing superpixel-guided clustering (SGC) to aggregate similar feature vectors, providing more accurate guidance. Although the above methods guide the prototypes from the support set, they encounter challenges due to the shot differences between the support set and query set. To address this issue, Fan et al. [35] introduced a novel matching strategy to alleviate the problem. This strategy involves using query prototypes to match the query features, where the query prototypes are collected from high-confidence query predictions. However, the limitation lies in the foreground prototypes, which are single prototypes that may not perform optimally for complex and diverse objects.

Despite the significant progress in FSS frameworks for natural image processing, their potential applications in other domains, such as remote sensing image interpretation, have remained largely unexplored. To bridge this gap, Yao et al. [36] proposed scale-aware modules that enable detailed matching of multiple prototypes, resulting in an excellent FSS performance for aerial images. Moreover, Lang et al. [37] addressed the challenges posed by the complex features in remote sensing images through progressive parsing and storing base class memories, effectively supporting their novel class prototype extraction method and achieving an outstanding performance. In addition, Lang et al. [38] proposed a novel and powerful remote sensing FSS framework with global rectification and decoupled registration for more tailored object localization. These works have greatly promoted the development of FSS

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

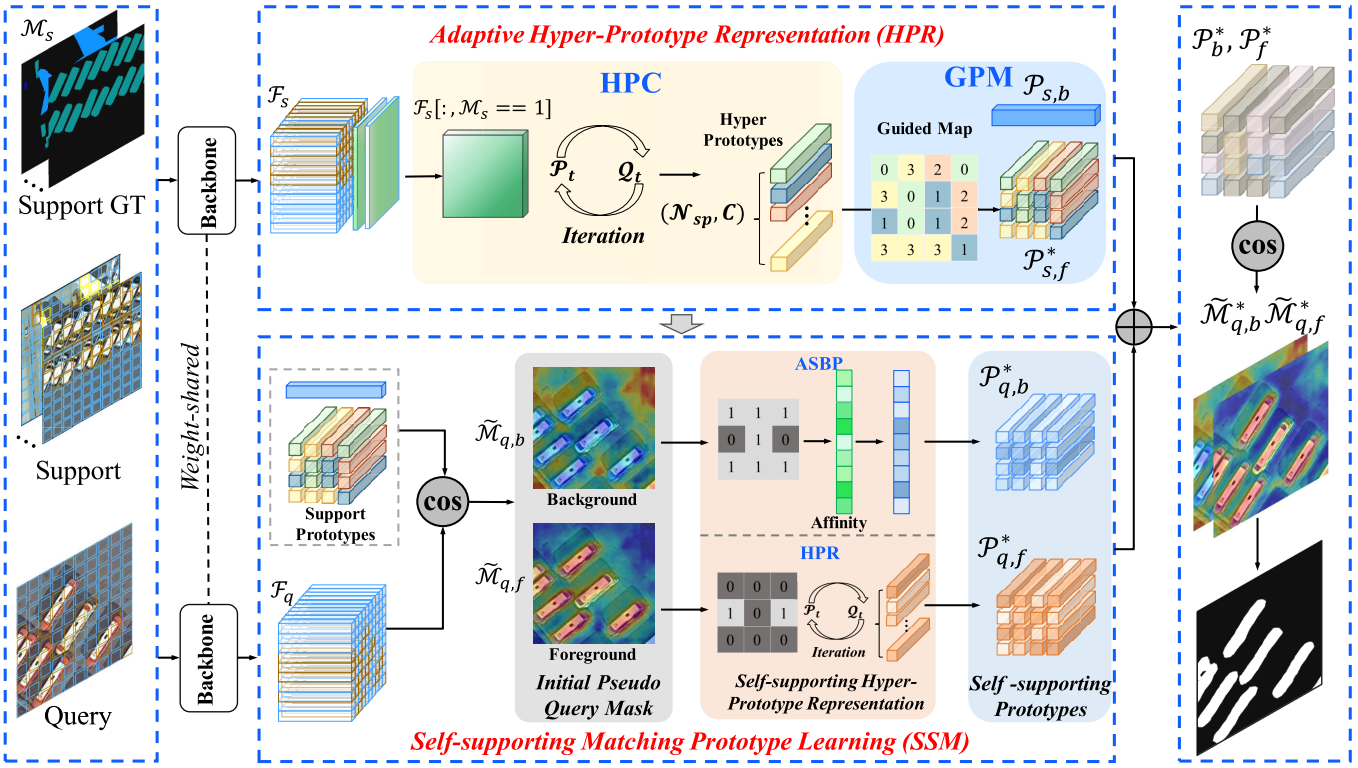



Fig. 2. Architecture illustration of the proposed SSMP, including the backbone network, HPC, GPM, and SSM. The support and query images are used to extract features. The HPC module is then used to extract the hyperprototypes of the support set, and GPM is used to generate the preliminary prediction result of the query set as a pseudo mask. The pseudo mask of the query set is then used to generate the query prototype by ASBP and HPR. All the prototypes are used to generate the final prediction result of the query set by similarity calculation.

in remote sensing. The have laid a strong foundation for our research, and have also shown that there is still great development potential for FSS in remote sensing. In view of the lack of research in the current field, we have meticulously crafted a comprehensive and innovative FSS framework, leveraging a combination of HPR and self-supporting matching prototype learning. This approach significantly enhances the quality of the extracted support and query prototypes by utilizing multiple prototypes from various sources (support and query) to effectively guide the interpretation of novel classes. We have seamlessly integrated these advancements into a unified few-shot framework called SSMP, offering a cohesive and powerful solution for elevating FSS to new heights.

## III. Methodology

### A. Task Description

In FSS, the goal is to enable the model to segment new classes with only a few annotated examples. Following the standard FSS settings, the task typically consists of two stages: a meta-training stage and a meta-testing stage. The dataset is divided into two parts—a training set $D_{\text{train}}$ and a test set $D_{\text{test}}$—and the corresponding semantic classes are $C_{\text{base}}$ and $C_{\text{novel}}$, respectively, where $C_{\text{base}} \cap C_{\text{novel}} = \varnothing$. Both $D_{\text{train}}$ and $D_{\text{test}}$ are composed of a large number of episodes $E_i = \{S, Q\}_{i=1}^{N}$, each of which contains a support set $S = \{(I_s^k, Y_s^k)\}_{k=1}^{K}$ and a query set $Q = \{(I_q, Y_q)\}$, totaling $K + 1$ image-label pairs. During the meta-training stage, we randomly sample a series of episodes $S$ from $D_{\text{train}}$, and the model is trained to the learn mapping function $f_\theta : \{(S, I_q)\} \rightarrow \{Y_q\}$ that can accurately segment the pixels belonging to the classes in $C_{\text{base}}$ given only a few support shots. In the meta-testing stage, the learned mapping function $f_\theta$ is evaluated in the corresponding regions in the query image $I_q$ sampled from $D_{\text{test}}$ with respect to the novel class support set $S$. The GT labels of the query mask $Y_q$ provided during the meta-training are not available during the meta-testing stage.

### B. Method Overview

In this article, to tackle the aforementioned problems, we propose an adaptive self-supporting prototype learning framework, namely SSMP, for more efficient and robust FSS. The overall architecture is shown in Fig. 2. SSMP contains three main components, which are described below: 1) adaptive self-supporting matching prototype learning from query set to query set to extract semantic prototypes that are closer to the target query set; 2) a multiprototype representation for hyperprototype clustering (HPC) to adaptively extract multiple prototypes of the target; and 3) supervised learning optimization strategy for self-supporting losses. SSMP is a fully end-to-end trainable framework for adaptive multiprototype learning, which can be trained by optimizing the lightweight and efficient multitask loss function.

### C. Adaptive HPR

*1) Hyperprototype Clustering:* The idea of HPC is inspired by the superpixel sampling network (SSN) [39], MaskSLIC

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

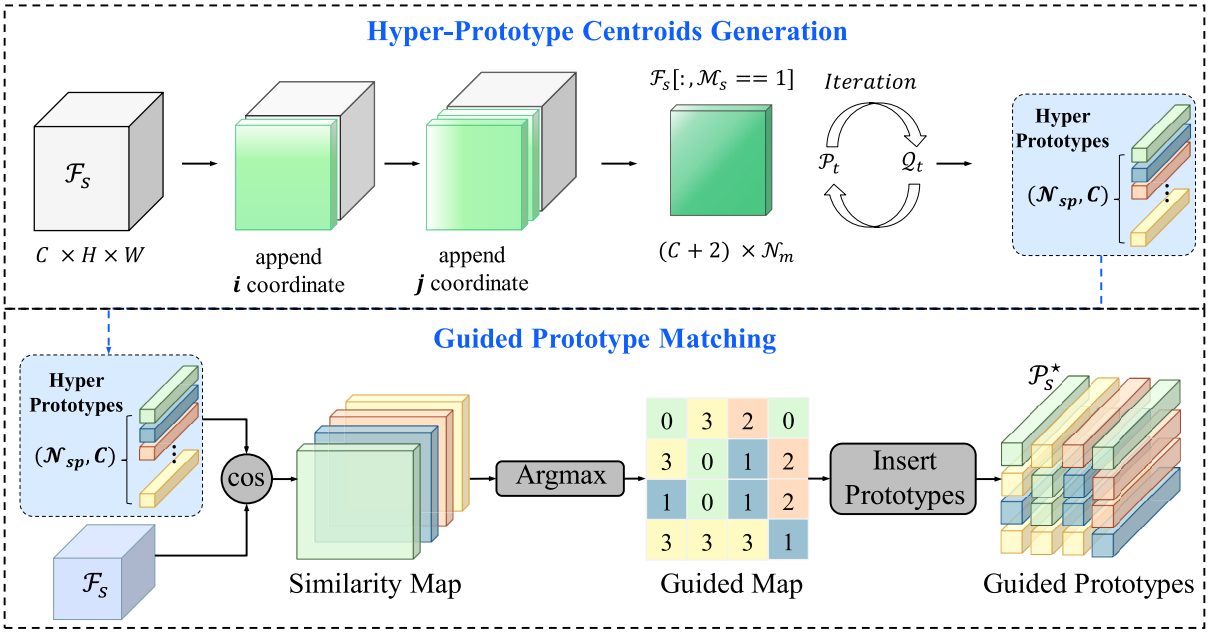



Fig. 3. Architecture illustration of HPR, including the hyperprototype centroids generation and the prototype allocation. The hyperprototypes of the support set are extracted using the HPR module. The cosine similarity between the hyperprototypes and the features are calculated to obtain the guided map, and then the prototypes are allocated to the query set by the guided map.

[40], and SGC [34]. The traditional superpixel-based lattice isoperimetric clustering (SLIC) [41] superpixel algorithm uses k-means clustering in two iterative steps: superpixel association of the superpixel centroids is updated iteratively. Based on the RGB similarity, the pixels are clustered into different superpixel centroids. Furthermore, the input image $I \in R^{(n+5)}$ is usually in a 5-D space (labxy) with n pixels, where lab denotes the pixel vector in the CIELAB color space, and xy denotes the pixel position. After iterative clustering, the algorithm outputs an association graph in which each pixel n is assigned to one of the superpixels.

Differing from using a single prototype to represent complex objects, this simple method aggregates the multiple hyperfeatures of the feature map extracted by the backbone by means of superpixel clustering. Since the hyperfeatures obtained by clustering from the feature map can be used as prototypes, called hyperprototypes, an obvious difference is that we are not clustering the hyperfeature in color space, but clustering the hyperprototypes in feature space.

The whole HPR process is shown in Fig. 3. Given the support features $F_s \in \mathbb{R}^{(c \times h \times w)}$, support set mask $M_s \in \mathbb{R}^{(h \times w)}$, and initial hyperprototype seed $P^0 \in \mathbb{R}^{(c \times N_{hp})}$ (where $N_{hp}$ is the number of hyperprototypes), the goal is to obtain the final centroids of hyperprototypes, which serve as multiple prototypes of the support set. First, we concatenate the spatial coordinates and feature maps of each pixel, introducing spatial position information. Then, after performing SLIC [41], a distance function $D$ is defined that consists of both the feature distance and the spatial distance

$$\mathcal{D} = \sqrt{d_f^2 + (d_s/r)^2} \tag{1}$$

where $d_f$ and $d_s$ are the Euclidean distance for the features and coordinate values, respectively, and $r$ is a weighting factor. The support mask is used to filter out the background information, keeping only the mask features, and the feature mapping is compressed from $F_s \in R^{c \times h \times w}$ to $F_s' \in R^{c \times N_m}$, where $N_m$ is the number of pixels inside the support mask.

Th hyperprototypes are then computed iteratively. For each iteration $t$, the association mapping $Q_t$ is first calculated between each feature $p$ and all the hyperprototypes based on the distance function $D$. $Q_t$ is calculated as follows:

$$\mathcal{Q}_{pi}^t = e^{-\mathcal{D}(\mathcal{F}_p', \mathcal{P}_i^{t-1})} = e^{-\|\mathcal{F}_p' - \mathcal{P}_i^{t-1}\|^2}. \tag{2}$$

The new hyperprototype centroids are then updated as the weighted sum of the masked features

$$\mathcal{P}_i^t = \frac{1}{\mathcal{Z}_i^t} \sum_{p=1}^{\mathcal{N}_m} \mathcal{Q}_{pi}^t \mathcal{F}_p' \tag{3}$$

where $\mathcal{Z}_i^t = \Sigma_p \mathcal{Q}_{pi}^t$ is a constant for normalization.

Next, we elaborate on the selection of the initial seeds. Typically, in hyperprototype algorithms, an $H \times W$ image is uniformly divided into regular grid units of size $h \times w$, and each grid unit is initialized as an initial seed (superfeature prototype). However, this general initialization is not suitable for our purpose of providing foreground mask support in the image. We only need to initialize seeds within the foreground region. To uniformly initialize seeds within the masked region, we refer to MaskSLIC [40] to iteratively place each initial seed (initial hyperprototype). This initialization strategy for the seeds enables faster and more stable convergence of the hyperprototypes through clustering, requiring only a few iterations.

*2) Adaptability-Guided Prototype Allocation:* In order to make the prototype matching more adaptive to the content of the query image, we use guided prototype matching (GPM), as shown in Fig. 3, which first computes the cosine distance to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

measure the similarity between each prototype and each query feature element

$$\mathcal{C}_i^{x,y} = \text{Cosine}\left(\mathcal{P}_i, \mathcal{F}_q^{x,y}\right) = \frac{\mathcal{P}_i \cdot \mathcal{F}_q^{x,y}}{\|\mathcal{P}_i\| \cdot \|\mathcal{F}_q^{x,y}\|} \quad i \in \{1, 2, \ldots, N_{hp} + 1\} \tag{4}$$

where $\mathcal{P}_i \in R^{c\times 1}$ is the $i$th hyperprototype centroid and background prototype, and $\mathcal{F}_q^{x,y} \in R^{c\times 1}$ is the feature vector at image location $(x, y)$ of the query feature. We calculate the cosine similarity between all the hyperprototypes and the target feature at each pixel location to find the hyperprototype that is most similar to the target feature, which is assigned to the query feature block, and each pixel location is matched to the hyperprototype that is most similar to the feature at the current location, as shown below

$$\mathcal{G}^{x,y} = \underset{i\in\{1,\ldots,\mathcal{N}_{hp}+1\}}{\text{argmax}} \ \mathcal{C}_i^{x,y}. \tag{5}$$

The allocation-guided hyperprototype block is obtained, and the final hyperprototype block consists of the hyperprototypes and the background prototypes obtained by masked average pooling (MAP).

### D. Self-Supporting Matching Prototype Learning

*1) Self-Supporting Prototypes:* Query prototypes are generated by aggregating the query features and are used to self-support the query features themselves. Recall that the regular support prototype generation process is

$$\mathcal{P}_{s,b} = \textbf{MAP}(\mathcal{M}_s, \mathcal{F}_s) \tag{6}$$

where MAP is the masked average pooling operation

$$\mathcal{M}_1 = \boldsymbol{softmax}\left(\boldsymbol{cosine}\left(\mathcal{P}_s, \mathcal{F}_q\right)\right) \tag{7}$$

where cosine is the cosine similarity metric.

The query prototype $\mathcal{P}_q$ is generated using a similar approach, except that the GT of the query image is not available in the inference phase. We therefore use the segmentation result $\hat{\mathcal{M}}_q$ of the initially predicted query image to aggregate the query features to generate the corresponding query prototype.

The initial prediction of the query image is assigned to the segmentation result by matching multiple prototypes. Multiple prototypes are clustered by the HPR module to generate multiple foreground prototypes $\mathcal{P}_i^t, \mathcal{P}_{s,f}^\star = \{\mathcal{P}_i^t\}, i \in \{1, 2, \ldots, N_{hp}\}$. The background prototypes $\mathcal{P}_{s,b}$ generated by MAP constitute the hyperprototype blocks which $\mathcal{P}_s = \{\mathcal{P}_{s,b}, \mathcal{P}_{s,f}^\star\}$ and the query features to perform most similar matching, to classify each pixel as foreground or background, obtaining $\hat{\mathcal{M}}_q$

$$\hat{\mathcal{M}}_q^{x,y} = \begin{cases} 0, & \mathcal{G}^{x,y} = N_{hp} + 1 \\ 1, & \mathcal{G}^{x,y} < N_{hp} + 1 \end{cases} \tag{8}$$

where $\mathcal{G}^{(x,y)}$ is the assigned hyperprototype ID, the $N_{hp} + 1$th hyperprototype is the background prototype, and the first $N_{hp}$ are multiple foreground hyperprototypes generated by HPR. $\hat{\mathcal{M}}_q^{x,y}$ is the value of $\hat{\mathcal{M}}_q$ at pixel coordinates of $(x, y)$, with 0 being the background and 1 being the foreground.

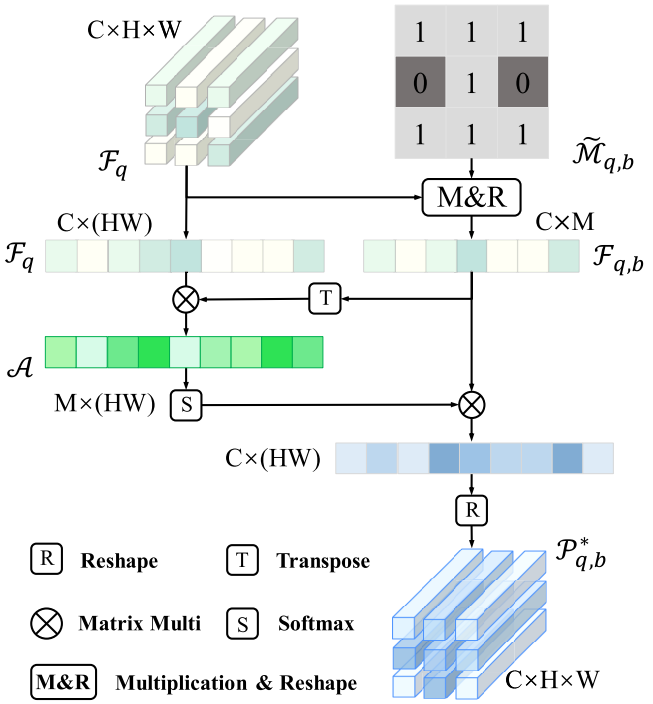



Fig. 4. Illustration of ASBP, including the background query feature aggregation, affinity matrix calculation, and ASBP generation.

*2) Adaptive Self-Supporting Background Prototype:* Fan et al. [35] pointed out that the background can be cluttered, and the commonality can be reduced to local semantic similarity in disjoint regions, sharing global semantic commonality among all the background pixels. For example, for a query image with airplane as the target class, other objects such as cars are considered as background, but they are different in appearance and semantic level. This observation is also validated by the lower background pixel similarity, compared to the foreground, especially in the case of intraobject images, thus generating multiple self-supporting background prototypes for different query semantic regions.

As shown in Fig. 4, ASBP is to dynamically aggregate similar background pixels for each query pixel to generate adaptive self-supporting background prototypes. Specifically, the background query feature $\mathcal{F}_q$ is first aggregated by multiplying the query feature $\mathcal{F}_q$ with the mask on the background mask $\hat{\mathcal{M}}_{q,b}$ to aggregate the background query feature $\mathcal{F}_{q,b} \in R^{c\times M}$, where this $M$ is the number of pixels in the background region. The affinity matrix $\mathcal{A}$ is computed by the matrix multiplication operation MatMul between the reconstructed background query feature $\mathcal{F}_{q,b}$ and the whole query feature $\mathcal{F}_q$. Calculation of $\mathcal{A}$ is as follows:

$$\mathcal{A} = \boldsymbol{MatMul}(\mathcal{F}_{q,b}^T, \mathcal{F}_q) \tag{9}$$

where $\mathcal{A}$ is of $\mathbb{R}^{M\times(HW)}$. The affinity matrix is normalized by a softmax operation along the first dimension, which is used to weight the aggregation of background query features for each query pixel to generate an adaptive self-supporting background prototype $\mathcal{P}_{q,b}^\star \in \mathbb{R}^{c\times h\times w}$

$$\mathcal{P}_{q,b}^\star = \boldsymbol{MatMul}(\mathcal{F}_{q,b}, \boldsymbol{softmax}(\mathcal{A})). \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

The self-supporting prototype is updated with the ASBP: $\mathcal{P}_q = \{\mathcal{P}^{\star}_{q,b}, \mathcal{P}^{\star}_{q,f}\}$.

### E. Prototype Matching

We weightedly combine the support prototype $\mathcal{P}_s$ and the self-supporting prototype $\mathcal{P}_q$ by weightedly fusing the background prototypes in the support prototype and the background prototypes in the self-supporting prototype $\mathcal{P}^{\star}_b$

$$\mathcal{P}^{\star}_b = \lambda_{s,b}\mathcal{P}_{s,b} \oplus \lambda_{q,b}\mathcal{P}^{\star}_{q,b}. \tag{11}$$

The corresponding foreground prototypes are fused to obtain $\mathcal{P}^{\star}_f$

$$\mathcal{P}^{\star}_f = \lambda_{s,f}\mathcal{P}^{\star}_{s,f} \oplus \lambda_{q,f}\mathcal{P}^{\star}_{q,f} \tag{12}$$

where $\lambda_{s,b}$, $\lambda_{s,f}$, $\lambda_{q,b}$, and $\lambda_{q,f}$ are the weighting coefficients of each prototype. The fused foreground and background prototypes are used as the final augmented prototype $\mathcal{P}^{\star}_s$

$$\mathcal{P}^{\star}_s = \{\mathcal{P}^{\star}_b, \mathcal{P}^{\star}_f\}. \tag{13}$$

We then compute the cosine distance between the augmented support prototype $\mathcal{P}^{\star}_s$ and query feature $\mathcal{F}_q$ to generate the final matching prediction

$$\mathcal{M}_2 = \boldsymbol{softmax}\big(\boldsymbol{cosine}\big(\mathcal{P}^{\star}_s, \mathcal{F}_q\big)\big). \tag{14}$$

The generated cosine distance maps are then supervised for training

$$\mathcal{L}_{\text{seg}} = \boldsymbol{BCE}(\boldsymbol{cosine}\big(\mathcal{P}^{\star}_s, \mathcal{F}_q\big), \mathcal{M}_q) \tag{15}$$

where $\boldsymbol{BCE}$ is the binary cross-entropy loss and $\mathcal{M}_q$ is the GT mask of the query image.

To further facilitate the self-supporting matching process, we propose a new pseudo-query supervised loss. The initial query prediction is obtained by calculating the cosine distance between the hyperprototypical block $\mathcal{P}_s = \{\mathcal{P}_{s,b}, \mathcal{P}_{s,f}\}$ and the query feature $\mathcal{F}_q$ for the support set $\hat{\mathcal{M}}_q$. This process is trained in a supervised manner with the loss function $\mathcal{L}_{\text{pseudo}}$

$$\mathcal{L}_{\text{pseudo}} = \boldsymbol{BCE}(\boldsymbol{cosine}(\mathcal{P}_{s,b} \oplus \mathcal{P}^{\star}_{s,f}, \mathcal{F}_q), \mathcal{M}_q). \tag{16}$$

We can apply the same procedure on the support feature to introduce the self-supporting matching loss $\mathcal{L}_{s-\text{hpr}}$

$$\mathcal{L}_{s-\text{hpr}} = \boldsymbol{BCE}(\boldsymbol{cosine}\big(\mathcal{P}_{s,b} \oplus \mathcal{P}^{\star}_{s,f}, \mathcal{F}_s\big), \mathcal{M}_s) \tag{17}$$

where $\mathcal{M}_s$ is the GT mask of the support image. Finally, we train the model in an end-to-end manner by jointly optimizing all the aforementioned losses

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{pseudo}} + \eta\mathcal{L}_{s-\text{hpr}} \tag{18}$$

where $\eta$ is a balancing coefficient set to 0.2.

## IV. Experiments

### A. Experimental Setup

*1) Datasets:* The existing available remote sensing image semantic segmentation datasets for the FSS task have two main drawbacks—the limited numbers and classes of semantic objects—which indicates that these datasets cannot satisfy the base requirement of the FSS task. Instead, the newly released iSAID [42], [43] dataset provides a feasible solution. The iSAID dataset contains 655 451 object instances for 15 categories across 2806 high-resolution images, which exactly matches the requirement of the FSS task. Based on the setup of PASCAL-5i [44], Yao et al. [36] designed an FSS benchmark iSAID-$5^i$ constructed from the large-scale iSAID dataset. The iSAID-$5^i$ [36] categories are divided into three category systems, namely split0, split1, and split2. Each category system contains five different categories, and the categories of the different category systems have no intersection. To further enlarge the available labeled data, we randomly cropped images of 256 × 256 from the original images to rebuild the iSAID-$5^i$ dataset. This dataset contains 18 076 images for training and 6363 images for validation. In the $k$-shot setting, $k + 1$ images would be randomly sampled from the training set $D_{\text{train}}$, and each image is guaranteed to have the same category label. Among the $k + 1$ images, $k$ images are input into the FSS model as support images accompanied by their annotated masks. The other image is regarded as the query image, and its annotation mask is regarded as the GT for calculating the supervision loss.

In order to properly evaluate the effectiveness of the proposed method, we needed to conduct experiments on more remote sensing FSS datasets. We referred to the processing of the iSAID-$5^i$ dataset for the land-cover dataset LoveDA [45] to obtain the LoveDA-$2^i$ dataset suitable for the FSS task. This dataset contains 22 698 images for training and 15 021 images for validation.

*2) Evaluation Metrics:* After reviewing the previous research in the field, we decided to utilize the mean intersection over union (mIoU) as the primary evaluation metric, alongside the foreground–background intersection over union (FB-IoU) as a secondary metric. These metrics can provide us with a comprehensive insight into the performance of the proposed approach. To be more specific, the equations for calculating these metrics are as follows: mIoU is calculated as the mIoU for each unseen class present in the target fold. It can be represented as $\mathbf{mIoU} = (1/n)\Sigma_{c=1}^{n}\mathbf{IoU}_c$, where n denotes the number of unseen classes in each target fold, and $\mathbf{IoU}_c$ represents the intersection over union for class c. FB-IoU, on the other hand, measures the IoU for both the foreground and background regions. It is computed as the average of the foreground IoU ($\mathbf{IoU}_F$) and the background IoU ($\mathbf{IoU}_B$), represented by the equation $\mathbf{FB\text{-}IoU} = 1/2(\mathbf{IoU}_F + \mathbf{IoU}_B)$. By employing these evaluation metrics, we can effectively assess the performance of the proposed method and compare it with previous works in a meaningful way.

*3) Implementation Details:* In order to enhance the generalization capability, we adopted a strategy based on the existing FSS models, leveraging pretrained backbone networks with frozen parameters. For comparison purposes, we utilized ResNet-50 [46] and ResNet-101 [46] as the backbone networks, initializing their weights using the pretrained weights from the base learner. The remaining layers are initialized according to the default settings of the PyTorch framework. To train our proposed few-shot model, we follow the meta-learning paradigm on the base dataset. The training process involves a batch size of 8 or 16 and runs for a total of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE RESULTS OF ONE-SHOT AND FIVE-SHOT SEGMENTATION ON iSAID-$5^i$ USING THE MIOU METRIC. "#PARAMS." INDICATES THE NUMBER OF PARAMETERS OF THE NETWORK, "#L-PARAMS" INDICATES THE NUMBER OF LEARNABLE PARAMETERS OF THE NETWORK. THE BEST PERFORMANCE IS SHOWN IN BOLD

| Backbone | Method | 1-shot | | | | 5-shot | | | | #L-Params | #Params |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Split-0 | Split-1 | Split-2 | Mean | Split-0 | Split-1 | Split-2 | Mean | | |
| ResNet50 | PANet (ICCV-19) [33] | 32.22 | 18.93 | 28.15 | 26.43 | 40.93 | 27.36 | 35.96 | 34.75 | 23.6M | 23.6M |
| | PFENet (TPAMI-20) [47] | 22.35 | 24.92 | 31.65 | 26.31 | 24.43 | 28.05 | 32.36 | 38.71 | 34.4M | 53.4M |
| | HSNet (ICCV-21) [48] | 28.49 | 24.66 | 39.36 | 30.84 | 36.64 | 29.87 | 40.11 | 35.54 | **2.6M** | 26.1M |
| | ASGNet (ICCV-21) [34] | 25.12 | 23.93 | 28.39 | 25.81 | 42.62 | 35.36 | 42.69 | 40.22 | 10.1M | 34.1M |
| | BAM (CVPR-22) [28] | 31.35 | 25.14 | 36.80 | 32.17 | 35.66 | 29.55 | 46.15 | 39.95 | 4.9M | 51.6M |
| | SSP (ECCV-22) [35] | 41.76 | 31.51 | 42.93 | 38.73 | 53.00 | 39.23 | 51.18 | 47.80 | 8.7M | **8.7M** |
| | SDM (TGRS-22) [36] | 32.32 | 23.38 | 36.47 | 30.72 | 37.64 | 27.37 | 45.63 | 36.88 | 19.6M | 19.6M |
| | R2Net (TGRS-23) [38] | 35.57 | 20.23 | 40.16 | 31.99 | 43.71 | 25.96 | 42.19 | 37.29 | 51.8M | 51.8M |
| | HDMNet (CVPR-23) [49] | 38.19 | 27.98 | 37.94 | 34.70 | 48.32 | 30.32 | 46.95 | 41.87 | 4.2M | 50.8M |
| | SSMP | **44.08** | **32.21** | **44.11** | **39.82** | **57.95** | **39.63** | **53.99** | **50.52** | 8.7M | **8.7M** |
| ResNet101 | PANet (ICCV-19) [33] | 28.72 | 17.64 | 18.62 | 28.72 | 41.56 | 27.11 | 37.51 | 35.39 | 42.6M | 42.6M |
| | PFENet (TPAMI-20) [28] | 19.50 | 23.63 | 30.16 | 24.43 | 19.73 | 26.23 | 27.50 | 24.48 | 53.4M | 53.4M |
| | HSNet (ICCV-21) [48] | 31.43 | 24.83 | 37.18 | 30.84 | 38.31 | 25.54 | 42.87 | 35.57 | **2.6M** | 45.1M |
| | ASGNet (ICCV-21) [34] | 23.99 | 24.57 | 27.09 | 25.22 | 24.11 | 24.43 | 26.78 | 25.11 | 10.4M | 53M |
| | BAM (CVPR-22) [28] | 32.15 | 28.89 | 33.81 | 31.61 | 42.51 | 31.97 | 47.29 | 40.59 | 4.9M | 70.6M |
| | SSP (ECCV-22) [35] | 41.20 | 30.87 | 42.55 | 38.21 | 53.08 | 39.56 | 49.70 | 47.45 | 27.7M | **27.7M** |
| | SDM (TGRS-22) [36] | 28.89 | 23.71 | 36.34 | 29.65 | 36.35 | 27.95 | 42.87 | 35.72 | 38.5M | 38.6M |
| | R2Net (TGRS-23) [38] | 32.36 | 26.65 | 41.81 | 33.61 | 45.67 | 29.75 | 46.22 | 40.55 | 70.8M | 70.8M |
| | HDMNet (CVPR-23) [49] | 31.97 | 24.05 | 36.94 | 30.99 | 39.43 | 29.38 | 37.75 | 35.52 | 4.2M | 69.9M |
| | SSMP | **43.87** | **31.64** | **45.86** | **40.45** | **53.66** | **39.69** | **52.33** | **48.56** | 27.7M | **27.7M** |

TABLE II

QUANTITATIVE RESULTS OF ONE-SHOT AND FIVE-SHOT SEGMENTATION ON LOVEDA-$2^i$ USING THE MIOU METRIC. "#PARAMS." INDICATES THE NUMBER OF PARAMETERS OF THE NETWORK, "#L-PARAMS" INDICATES THE NUMBER OF LEARNABLE PARAMETERS OF THE NETWORK. THE BEST PERFORMANCE IS SHOWN IN BOLD

| Backbone | Method | 1-shot | | | | 5-shot | | | | #L-Params | #Params |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Split-0 | Split-1 | Split-2 | Mean | Split-0 | Split-1 | Split-2 | Mean | | |
| ResNet50 | PANet (ICCV-19) [33] | 10.68 | 9.80 | 6.74 | 9.07 | 17.51 | 13.32 | 12.18 | 14.34 | 23.6M | 23.6M |
| | PFENet (TPAMI-20) [47] | 16.09 | 19.56 | 17.07 | 17.57 | 17.36 | 22.51 | 19.26 | 19.71 | 34.4M | 53.4M |
| | HSNet (ICCV-21) [48] | 15.30 | 25.91 | 29.87 | 23.69 | 17.65 | 29.62 | 35.79 | 27.69 | **2.6M** | 26.1M |
| | ASGNet (ICCV-21) [34] | 16.13 | 16.48 | 14.11 | 15.57 | 17.62 | 23.01 | 32.39 | 24.34 | 10.1M | 34.1M |
| | BAM (CVPR-22) [28] | 14.13 | 14.37 | 17.68 | 15.39 | 10.42 | 17.51 | 26.15 | 18.03 | 4.9M | 51.6M |
| | SSP (ECCV-22) [35] | 19.57 | 21.98 | 28.67 | 23.41 | 31.38 | 28.68 | 36.52 | 32.19 | 8.7M | **8.7M** |
| | SDM (TGRS-22) [36] | 17.53 | 21.79 | 29.29 | 22.87 | 18.58 | 25.92 | 33.75 | 26.08 | 19.6M | 19.6M |
| | R2Net (TGRS-23) [38] | 23.62 | **32.39** | 35.92 | **30.64** | 24.79 | 35.94 | 37.92 | 32.88 | 51.8M | 51.8M |
| | HDMNet (CVPR-23) [49] | 20.23 | 22.82 | 21.31 | 21.45 | 23.13 | 30.97 | 22.60 | 25.57 | 4.2M | 50.8M |
| | SSMP | **25.33** | 25.82 | **36.51** | 29.22 | **33.16** | **36.51** | **42.62** | **37.43** | 8.7M | **8.7M** |
| ResNet101 | PANet (ICCV-19) [33] | 10.99 | 9.42 | 6.75 | 9.05 | 18.08 | 16.88 | 11.26 | 15.41 | 42.6M | 42.6M |
| | PFENet (TPAMI-20) [28] | 16.44 | 19.71 | 16.18 | 17.44 | 16.79 | 22.33 | 14.92 | 18.01 | 53.4M | 53.4M |
| | HSNet (ICCV-21) [48] | 15.03 | 24.34 | 30.85 | 23.41 | 16.40 | 26.87 | 37.12 | 26.80 | **2.6M** | 45.1M |
| | ASGNet (ICCV-21) [34] | 12.67 | 15.22 | 16.18 | 14.69 | 16.34 | 13.58 | 14.84 | 14.92 | 10.4M | 53M |
| | BAM (CVPR-22) [28] | 6.93 | 13.78 | 27.26 | 15.99 | 15.32 | 20.45 | 33.02 | 22.93 | 4.9M | 70.6M |
| | SSP (ECCV-22) [35] | 19.69 | 20.14 | 26.54 | 22.12 | 27.40 | 35.70 | 35.04 | 32.71 | 27.7M | **27.7M** |
| | SDM (TGRS-22) [36] | 17.85 | 22.01 | 29.96 | 23.27 | 18.53 | 24.73 | 33.39 | 25.55 | 38.5M | 38.6M |
| | R2Net (TGRS-23) [38] | 23.23 | **30.50** | 35.57 | 29.77 | 24.33 | **36.90** | 31.66 | 30.96 | 70.8M | 70.8M |
| | HDMNet (CVPR-23) [49] | 16.01 | 20.27 | 16.07 | 17.45 | 21.59 | 25.82 | 23.45 | 23.62 | 4.2M | 69.9M |
| | SSMP | **25.11** | 27.64 | **36.76** | **29.84** | **30.41** | 35.80 | **42.62** | **36.28** | 27.7M | **27.7M** |

20 epochs. We employ the stochastic gradient descent (SGD) optimizer with momentum set to 0.9 and weight decay set to 0.0005. To ensure consistency, we preprocess the raw images from the iSAID dataset by cropping them to a size of 256 × 256 before feeding them into the model during training. For the meta-testing phase, we randomly selected 1000 episodes to evaluate the model's performance. The reported results are the averaged outcomes of five independent runs, each with a different seed. Our network was implemented using the PyTorch framework, and all the experiments were conducted on NVIDIA GeForce RTX 3090 GPUs.

### *B. Comparison With the State-of-the-Art*

In this section, we comprehensively compare the proposed model with the state-of-the-art FSS approaches. The model is evaluated from quantitative and qualitative perspectives to demonstrate its superiority. Tables I and II present the mIoU results of the different methods under one-shot and five-shot settings on the iSAID-$5^i$ and LoveDA-$2^i$ datasets, respectively. The proposed SSMP method achieves the highest overall mIoU score in almost all cases, demonstrating its exceptional performance. On the iSAID-$5^i$ dataset, SSMP achieves the best performance in the different dataset splits as well as in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

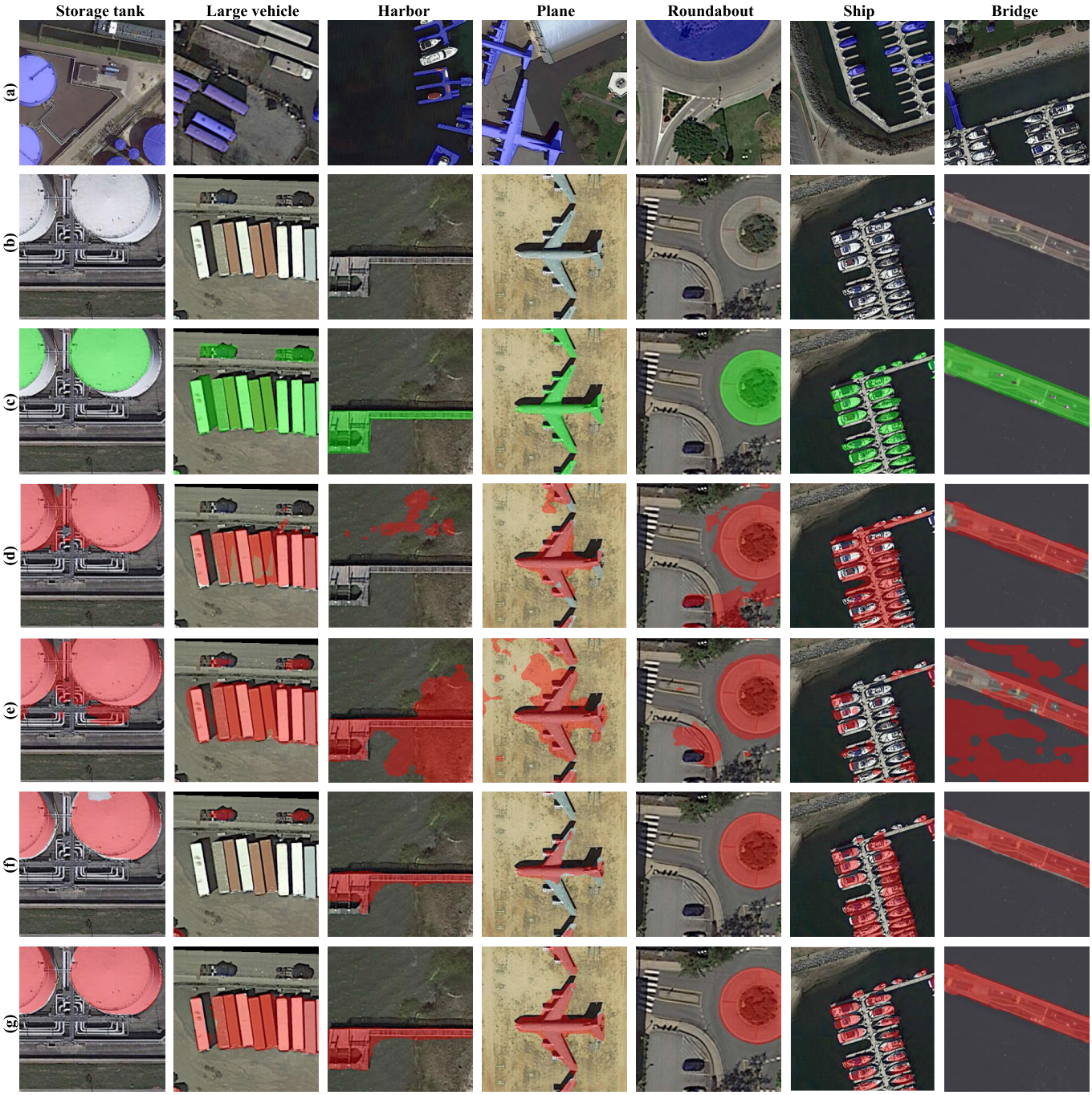



Fig. 5. Qualitative results of five-shot segmentation on the iSAID-$5^i$ dataset. (From Top to Bottom) Support images with GT marked as blue, the query images, the query GT marked as green, the predictions of the SDM, R2Net, and SSP approach, and the predictions (red masks) of the SSMP approach respectively. (From Left to Right) Storage tank, large vehicle, harbor, plane, roundabout, ship, and bridge, respectively. (a) Support. (b) Query. (c) Query GT. (d) SDM. (e) R2Net. (f) SSP. (g) Ours.

the different shot settings, where the SSP and SSMP methods significantly outperform the other FSS comparison methods. Specifically, when employing ResNet-50 and ResNet-101 as backbone networks, the proposed approach achieves performance gains over the state-of-the-art FSS model of SSP of 2.72% and 1.02% in the five-shot setting and 1.09% and 2.24% in the one-shot setting, respectively. On the remote sensing land-cover dataset LoveDA-$2^i$, SSMP and the state-of-the-art FSS model of R2Net [38] outperform the other comparison methods by a wide margin. SSMP significantly outperforms R2Net [38] in the five-shot setting and slightly outperforms it in the one-shot setting in the other category systems, despite being inferior to it in the split-1 category system. Overall, these results underscore the unique advantages of the proposed approach in addressing the current task. Furthermore, we note that the improved strategies do not introduce many additional learnable parameters, remaining comparable to the advanced methods. This indicates that the proposed approach achieves an excellent performance while maintaining model simplicity. The effectiveness of the proposed framework is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

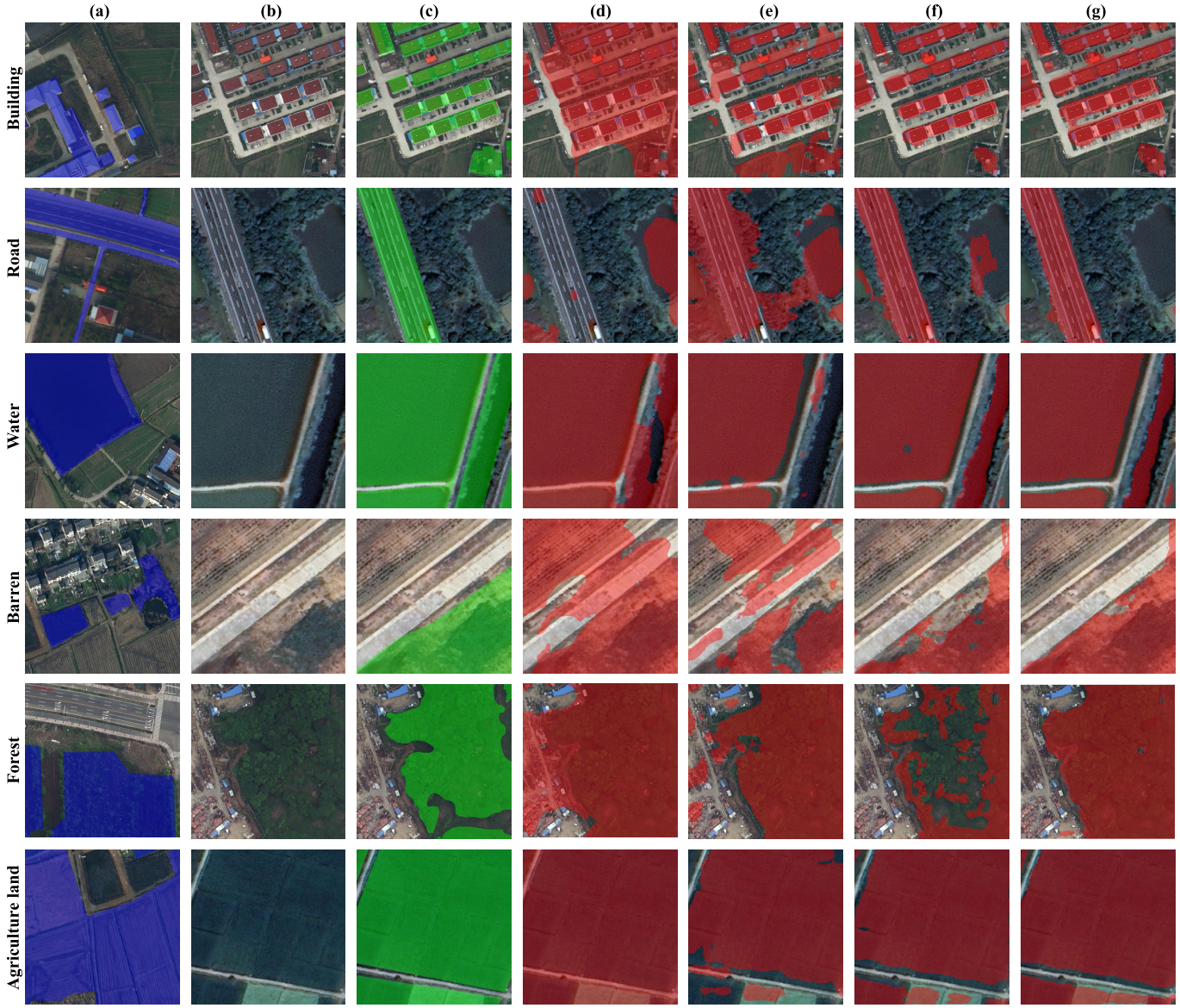



Fig. 6. Qualitative results of five-shot segmentation on the LoveDA-$2^i$ dataset. (From Left to Right) Support images with GT marked as blue, the query images, the query GT marked as green, the predictions of the SDM, R2Net, and SSP approach, and the predictions (red masks) of the SSMP approach respectively. (From Top to Bottom) Building, road, water, barren, forest, and agriculture land, respectively. (a) Support. (b) Query. (c) Query GT. (d) SDM. (e) R2Net. (f) SSP. (g) Ours.

further illustrated by the FB-IoU results, as shown in Table III. Through rigorous analysis, both quantitative and qualitative, we have substantiated the superior performance of the proposed model compared to the state-of-the-art FSS approaches.

Fig. 5 shows the qualitative results of the proposed approach and the other state-of-the-art FSS models of SSP [35], SDM [36], and R2Net [38] under the five-shot setting. The first row in Fig. 5(a) is the support set of different land objects, where the blue area is the mask of the categories in the support set. The second row in Fig. 5(b) is the query set, i.e., the image that the model is going to align the pixel set segmentation, and the third row in Fig. 5(c) is the GT of the query set. The following four rows in Fig. 5(d)–(g) show the segmentation results of the representative comparison methods and the proposed SSMP method on the query set of different land objects. It can be observed that incomplete segmented objects and irrelevant background objects appear in the segmentation results of the SSP [35] approach, and the results for dense small objects such as boats and cars show object sticking. R2Net [38] makes serious errors in segmenting features with obscure geographic boundaries, such as harbors, while SDM [36] is unable to perceive the harbor outlook effectively and thus segments it incorrectly. In contrast, the proposed approach can better segment intact objects with blurred geographic boundaries, such as harbors, guided by the query set. It can also distinguish the dense small objects, to some extent, as shown in the results for boats.

As shown in Fig. 6, we visualized the six land-cover classes of the LoveDA dataset: building, road, water, barren, forest, and agriculture land. For dense buildings, our method and SSP achieve relatively good segmentation results. However, our method excels in accurately segmenting buildings with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

ambiguous boundaries. In contrast, R2Net and SDM methods tend to erroneously segment features such as agriculture land and roads as buildings. Regarding roads, the SDM method fails to correctly segment them, while R2Net and SSP suffer from significant segmentation errors, incorrectly segmenting forests and water as roads. Our approach accurately identifies road segments. For water, SDM and R2Net exhibit segmentation errors and instances of undersegmentation, while SSP and our method achieve relatively accurate segmentation of water bodies. However, SSP shows minor cases of omission, and our method has room for improvement in boundary refinement. Regarding extensive features like barren and forest with indistinct boundaries, SDM and R2Net incorrectly segment some background as target features. SSP performs well in barren segmentation but experiences significant undersegmentation in forest areas, possibly due to texture variations within forested regions. Our method provides more accurate segmentation for barren and forest features. For agriculture land, SDM incorrectly classifies field tracks as part of the agriculture land, R2Net struggles to accurately segment agriculture land with significant texture variations, and SSP occasionally misses segments. In contrast, our method accurately segments agriculture land. Overall, our method demonstrates strong segmentation performance on remote sensing land-cover datasets, validating its effectiveness.

In addition, the SSMP approach can be flexibly extended to the multiclass few-shot semantic segmentation task, and the SSP [35] model, which is a representative and powerful model, can also implement the few-shot semantic segmentation of multiclass targets to compare with SSMP. The datasets for the multiclass FSS task were based on iSAID-$5^i$ and LoveDA-$2^i$ with the images in which multiple target categories co-occur filtered out, e.g., under the split-2 category system of the LoveDA-$2^i$ dataset, both forest and agriculture land categories appear in the images. As shown in Table IV, the proposed method outperforms SSP [35] in terms of the mIoU in both the one-shot and five-shot settings. Overall, the experiments prove that the proposed SSMP framework is superior to the other FSS methods in both the traditional single-class and multiclass FSS task.

### C. Ablation Studies

We employed the ResNet-50 [46] backbone network for conducting ablation studies in the FSS task. Unless stated otherwise, the results presented primarily consist of average mIoU metrics obtained from a one-shot evaluation and three splits within a five-shot configuration.

*1) Effects of the Proposed Modules:* We first performed a detailed analysis of the proposed modules to evaluate their impact on model performance. As shown in Table V, the ultimate goal of the self-supporting matching (SSM) and HPR modules and the refine operation is to provide a more representative and robust prototype representation of the query set. However, their functional focuses are different, in that HPR focuses on generating multiple prototypes to represent complex feature objects, while SSM focuses on generating query pseudo labels to generate important query set prototypes. The two modules can also be flexibly used in conjunction with each other. The refine operation denotes the refinement of the self-supporting prototype by segmenting the initial query pseudo mask extracted by SSM again to obtain the secondary pseudo labels, and thus obtaining the prototype of the query set based on the more robust secondary pseudo labels. From the quantitative results, it can be observed that both modules and the refine operation consistently boost the accuracy of the few-shot model in all cases, and the refine operation with the HPR and SSM modules achieves the best performance, i.e., 39.82% in the one-shot setting and 50.52% in the five-shot setting.

TABLE III

PERFORMANCE COMPARISON ON iSAID-$5^i$ AND LoveDA-$2^i$ IN FB-IoU METRIC UNDER ONE-SHOT AND FIVE-SHOT SETTINGS. Δ INDICATES THE CHANGE IN PERFORMANCE FROM ONE-SHOT TO FIVE-SHOT

| Backbone | Method | iSAID-$5^i$ (FB-IoU) | | | LoveDA-$2^i$ (FB-IoU) | | |
|---|---|---|---|---|---|---|---|
| | | 1-shot | 5-shot | Δ | 1-shot | 5-shot | Δ |
| ResNet50 | PANet [33] | 50.67 | 57.15 | 6.48 | 45.66 | 49.73 | 4.07 |
| | PFENet [47] | 50.59 | 48.91 | -1.68 | 34.32 | 34.01 | -0.31 |
| | HSNet [48] | 54.72 | 57.49 | 2.77 | 45.31 | 43.39 | -1.92 |
| | ASGNet [34] | 49.84 | 57.85 | 8.01 | 35.41 | 34.14 | -1.26 |
| | BAM [28] | 56.57 | 58.25 | 1.68 | 38.62 | 37.82 | -0.80 |
| | SSP [35] | 60.52 | 66.12 | 5.60 | 43.44 | 51.09 | 7.66 |
| | SDM [36] | 53.03 | 57.10 | 4.07 | 38.48 | 35.60 | -2.87 |
| | R2Net [38] | 48.14 | 55.29 | 7.15 | 45.51 | 49.58 | 4.07 |
| | HDMNet [49] | 55.31 | 59.07 | 3.76 | 41.05 | 48.35 | 7.30 |
| | SSMP | 56.88 | 65.94 | 9.06 | 47.93 | 53.37 | 5.43 |
| ResNet101 | PANet [33] | 52.14 | 60.48 | 8.34 | 45.34 | 50.79 | 5.44 |
| | PFENet [47] | 48.45 | 48.59 | 0.14 | 36.82 | 31.92 | -4.90 |
| | HSNet [48] | 55.77 | 58.85 | 3.08 | 41.53 | 39.61 | -1.92 |
| | ASGNet [34] | 48.90 | 48.68 | -0.22 | 38.72 | 34.32 | -4.40 |
| | BAM [28] | 53.08 | 59.08 | 6.00 | 32.43 | 41.92 | 9.55 |
| | SSP [35] | 59.86 | 66.22 | 6.36 | 42.05 | 50.61 | 8.56 |
| | SDM [36] | 52.46 | 55.79 | 3.33 | 32.63 | 34.71 | 2.08 |
| | R2Net [38] | 50.65 | 58.76 | 8.11 | 44.62 | 50.64 | 6.02 |
| | HDMNet [49] | 51.84 | 57.95 | 6.09 | 40.27 | 44.76 | 4.49 |
| | SSMP | 59.80 | 65.45 | 5.65 | 48.56 | 50.89 | 2.34 |

TABLE IV

PERFORMANCE COMPARISON IN MULTICLASS FEW-SHOT SEMANTIC SEGMENTATION UNDER ONE-SHOT AND FIVE-SHOT SETTINGS. Δ INDICATES THE PERFORMANCE GAIN FROM ONE-SHOT TO FIVE-SHOT

| Dataset | Backbone | Method | M-IoU | | Δ |
|---|---|---|---|---|---|
| | | | 1-shot | 5-shot | |
| iSAID | ResNet50 | SSP [35] | 25.45 | 37.87 | 12.42 |
| | | SSMP | 27.19 | 41.03 | 13.85 |
| | ResNet101 | SSP [35] | 28.04 | 39.25 | 11.21 |
| | | SSMP | 29.49 | 39.36 | 9.87 |
| LoveDA | ResNet50 | SSP [35] | 18.91 | 23.88 | 4.97 |
| | | SSMP | 23.19 | 31.36 | 8.17 |
| | ResNet101 | SSP [35] | 17.59 | 26.44 | 8.85 |
| | | SSMP | 21.29 | 29.84 | 8.56 |

TABLE V

ABLATION STUDY OF THE MAIN MODEL COMPONENTS

| HPR | SSM | Refine | Split-0 | Split-1 | Split-2 | Mean |
|---|---|---|---|---|---|---|
| ✗ | ✗ | ✗ | 47.53 | 37.14 | 46.89 | 43.85 |
| ✗ | ✓ | ✗ | 52.00 | 39.31 | 49.04 | $46.78_{\uparrow 2.93}$ |
| ✗ | ✓ | ✓ | 53.00 | 39.23 | 51.18 | $47.80_{\uparrow 3.95}$ |
| ✓ | ✓ | ✗ | 53.58 | 38.9 | 51.12 | $47.87_{\uparrow 4.02}$ |
| ✓ | ✓ | ✓ | 57.95 | 39.17 | 53.99 | $50.37_{\uparrow 6.52}$ |

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

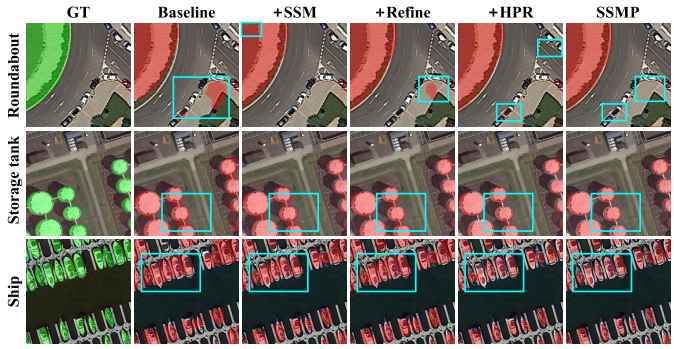



Fig. 7. Qualitative results of the different proposed modules. The baseline model and the models composed of different modules are used to visualize and analyze the inference results. The green area is the GT, the red area is the prediction results, and the cyan border is the focus area for analyzing the effect of the modules.

Visualization of the results of the ablation experiments combined with qualitative data is necessary to better understand the advantages offered by the different modules. Fig. 7 shows the visualization of the experimental results of the models composed of different modules according to Table V of the ablation experiments. The first column is the query set, where the green area is the query GT; the second column is the result of the baseline model; the third to fifth columns are the visualization results of adding the SSM module, the refine operation, and the HPR module to the baseline model; and the last column is the qualitative visualization result of the proposed SSMP model on three types of objects. It can be observed that there is target mis-segmentation in the baseline result, and there is severe segmentation target sticking in the dense target extraction. When we add the SSM module, the mIoU is improved by 2.93%, and the phenomenon of mis-segmentation of background into targets is alleviated, but the problem of dense target sticking is even more severe. This shows that the SSM module enhances the perception of foreground targets, to a certain extent. The mIoU is improved by 3.95% with the addition of the refine operation, which alleviates some of the dense target sticking, but then mis-segmentation of the feature targets occurs again. By introducing the HPR module, the mIoU is improved by 4.02% and the target mis-segmentation problem is significantly alleviated, as well as the dense target sticking. From the final SSMP model segmentation results, it can be observed that the model can correctly perceive the foreground feature targets, and the phenomenon of dense target sticking is almost nonexistent.

*2) Analysis of the Adaptive HPR:*

*a) Number of pixels for each hyperprototype:* The number of pixels $N_p$ for the HPC centroids is the number of pixels for each hyperprototype in the HPR module. When $N_p$ is smaller, the maximum count of hyperprototypes is larger. Therefore, to explore the effect of the hyperprototype count on the performance of the model, we conducted experiments to evaluate the impact of the $N_p$ on the performance of the model. The results are shown in Fig. 8. At the one-shot setting, the model performance fluctuates greatly with the number of pixel values and is not very stable, while at the five-shot setting, the model performance is similar for similar numbers of pixels. It can be easily observed that the model achieves the best performance in the five-shot settings when the number of pixels is about 50, but the best performance in the one-shot setting is when the number of pixels is {50, 140}.

*b) Analysis of the initial seeds in the HPR module:* To uniformly initialize the seeds in the masked region, we referred to MaskSLIC [40] for iteratively placing each initial seed. The pipeline of the seed growth is depicted in Fig. 9. It can be observed that the numbers of initial seeds are not the same for features of different sizes and complexity. Complex features, such as tennis courts and baseball diamonds with a high pixel ratio, have a higher number of iterations and more initial seeds. Meanwhile, the opposite is true for simple low pixel ratio features, such as boats and small airplanes. The number of seeds set in the initial iteration also determines the number of hyperprototype centroids of mass obtained from the final HPR module. This seed initialization results in faster convergence of the hyperprototype learning in only a few iterations.

*3) Ablation Study for the Self-Supporting Matching Prototype Learning:*

*a) Effects of the mask threshold:* The mask threshold $\tau$ includes the threshold of the foreground mask and the background mask in the SSM module. It influences the performance of the query pseudo labels in the SSM module. We conducted experiments to evaluate the impact of the mask threshold on the performance of the model. We set the foreground and background threshold to {0.5, 0.6, 0.7, 0.8, 0.9}. Two-by-two combinations of the foreground and background thresholds were applied in the ablation experiments. The results of the ablation study are shown in Fig. 10. It can be observed that the model achieves a good performance in the one-shot setting when the foreground threshold $\tau_{fg} \in [0.5, 0.6]$ and the background threshold $\tau_{bg} \in [0.5, 0.7]$. The best performance in the five-shot setting is when $\tau_{fg} = 0.8$ and $\tau_{bg} = 0.7$. It can be observed from the heat maps under the two settings that, as a whole, the performance of the one-shot setting model is affected by the fact that it is more sensitive to the threshold, while the five-shot setting model achieves a good performance over a wide range of thresholds.

*b) Separability analysis for the hyperprototypes:* Multiple hyperprototypes can be obtained through HPC for guiding the matching of the query sets. In order to determine the difference between the multiple prototypes obtained by the HPR module and the single prototypes obtained with the traditional MAP, the hyperprototypes generated by the HPR module and the single prototypes generated by MAP during the model inference process are visualized using the t-SNE visualization technique. Fig. 11(a)–(c) shows the single prototypes obtained by MAP, and Fig. 11(d)–(f) shows the multiple supporting hyperprototypes obtained by the HPR module for each support set. From this, it can be observed that the total number of prototypes obtained by MAP is small, and the categorization boundaries of the similar categories are less obvious. The total number of prototypes obtained by the HPR module is large, and the same support set generates multiple prototypes that better represent the category and are easily clustered.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

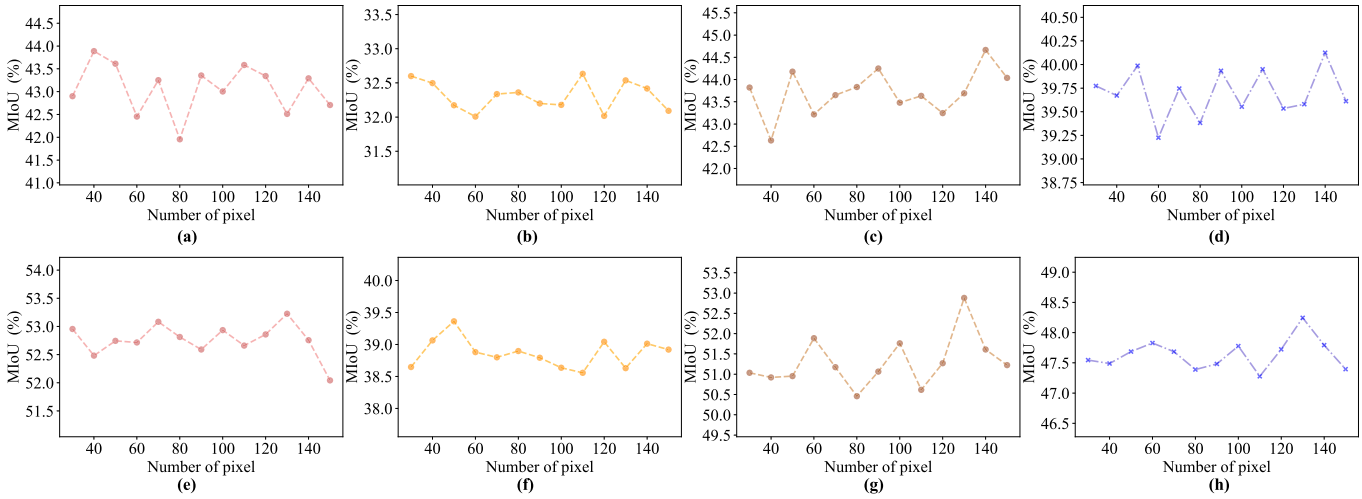



Fig. 8. Ablations studies of the number of pixels for each hyperprototype in different splits and shots. (a) Fold-0 in one-shot settings. (b) Fold-1 in one-shot settings. (c) Fold-2 in one-shot settings. (d) and (h) Average performance of all the splits corresponding to one-shot and five-shot settings. (e) Fold-0 in five-shot settings. (f) Fold-1 in five-shot settings. (g) Fold-2 in five-shot settings.

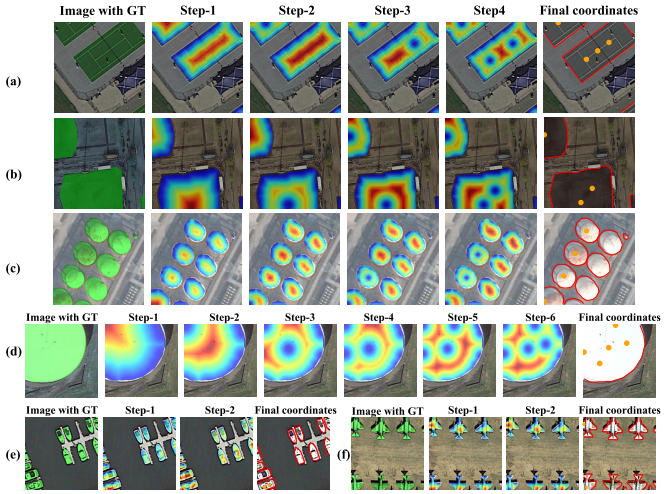



Fig. 9. Qualitative results of the seed growth. (1) Step 1: Perform a distance transformation and select the maximum point. (2) Step 2: Ignore the point and set it to 0, then repeat step 1. After this, iterate step 2 until all the coordinates of all the initial seeds are obtained. (a)–(f) Seed growth for different scenario objects.

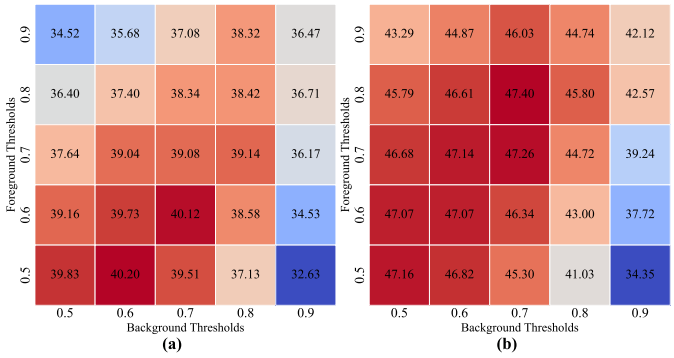



Fig. 10. Results of mask threshold variations for the self-supporting prototypes in one-shot and five-shot settings. (a) Average in one-shot settings. (b) Average in five-shot settings.

*c) Effects of the balancing coefficient $\eta$:* $\eta$ is a balancing coefficient introduced in the prototype matching to balance the optimization of the self-supporting matching loss with the segmentation loss function. In order to explore the effect of $\eta$ on the model performance, we carried out a series of ablation experiments on it. As shown in Table VI, the experimental range of $\eta$ is {0.1 - 0.5}, which shows the average results of the mIoU of the three-split category systems for the iSAID-5i dataset under the one-shot and five-shot setting. The "Mean" is the average result of the one-shot and five-shot settings, from which it can be observed that the best model performance is with $\eta$ set to 0.2. A reasonable $\eta$ setting can maximize the advantage of the self-supporting matching loss. Meanwhile, overall, the model's performance does not differ much with different $\eta$ settings, indicating that the model has strong robustness.

TABLE VI
ABLATION STUDY FOR THE BALANCING COEFFICIENT $\eta$

| $\eta$ | 1-shot | 5-shot | Mean |
|---|---|---|---|
| 0.1 | 38.89 | 48.18 | 43.54 |
| 0.2 | **39.89** | **48.33** | **44.11** |
| 0.3 | 39.51 | 47.67 | 43.60 |
| 0.4 | 39.83 | 47.67 | 43.75 |
| 0.5 | 39.02 | 47.19 | 43.11 |

*d) Effects of the number of test categories:* In order to explore the extent to which changes in the number of categories in the test set and the number of categories in the training set affect the performance of the model on the iSAID dataset, we evaluated the performance of the model by continuously increasing the number of categories in the test set, $N_{\text{test}}$, while decreasing the number of categories in the training set, $N_{\text{train}}$, $N_{\text{test}} + N_{\text{train}} = 15$. As shown in Fig. 12, when the number of categories in the test set is increased while the corresponding number of categories in the training set is decreased, the model's evaluation accuracy in the test set generally tends to decrease. The reduction in the number of training categories weakens the model's ability, while the difficulty of segmenting the new categories leads to a decrease in test accuracy.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

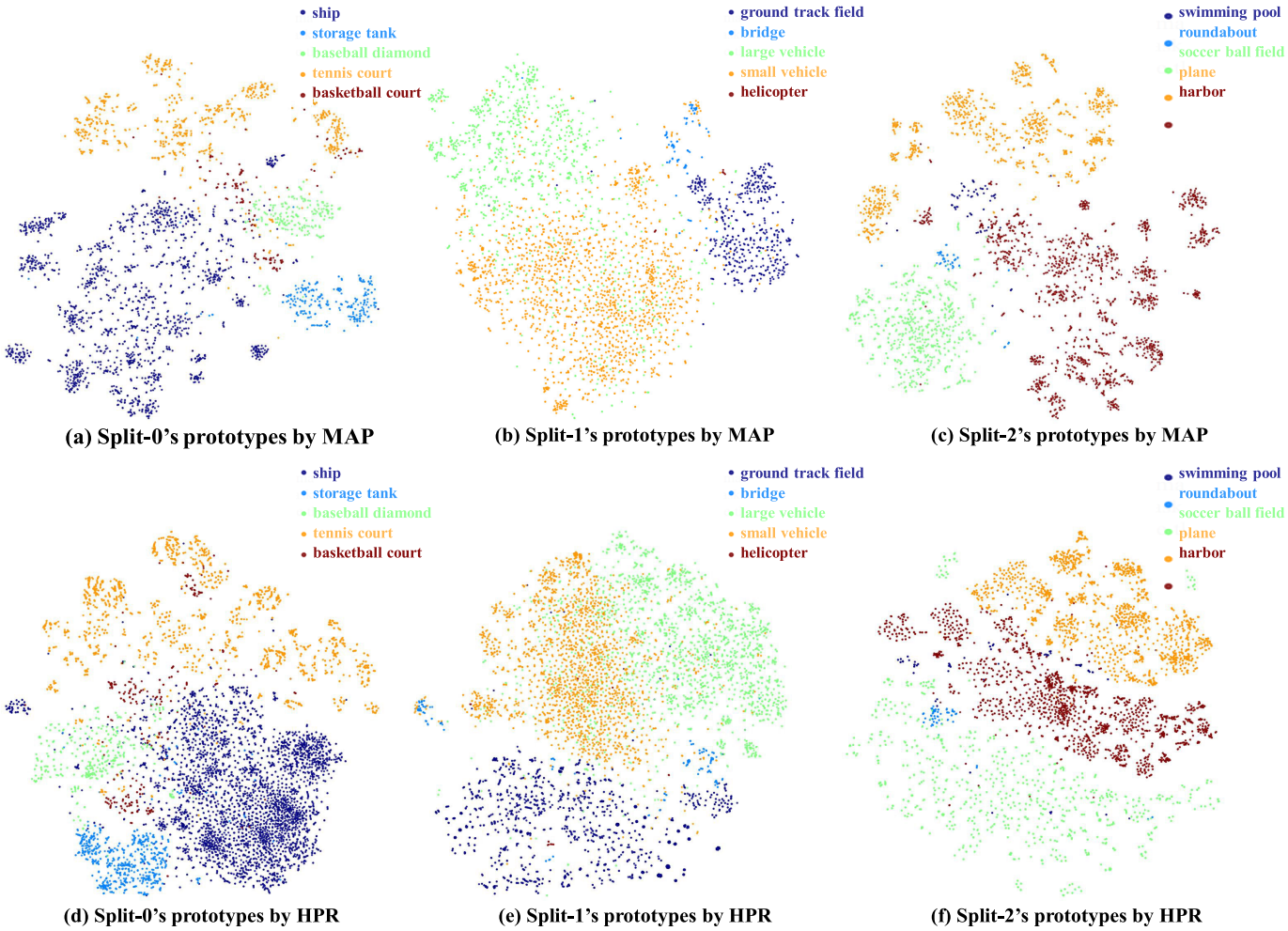



Fig. 11. Quantitative t-SNE results for the prototypes generated during the inference process. (a)–(c) Single prototypes generated by MAP for the support features. (d)–(f) Multiple hyperprototypes generated by the HPR module.

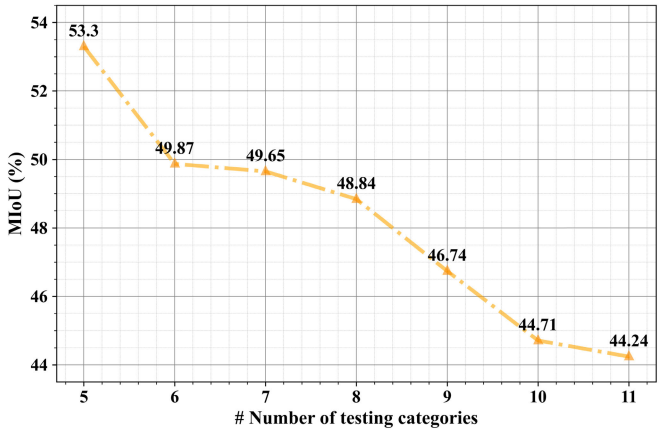



Fig. 12. Trend analysis for changes in the number of test categories.

## V. Conclusion

The diversity of remote sensing images and the complexity of feature objects cause inconvenience in the study of semantic segmentation tasks with low data volumes. Therefore, we propose adaptive HPR and self-supporting matching prototype learning for the few-shot framework of remote sensing images, which is referred to as SSMP. The adaptive HPR takes the foreground features corresponding to the support image to obtain the superpixel centroids, which results in multiple important foreground support prototypes. The self-supporting matching prototype learning further guides the prediction of the query sets by utilizing the support prototypes extracted in the previous phase to predict the query sets and obtain preliminary query set prediction results, which are used as pseudo labels for the query set prototype extraction. The proposed SSMP method has a variety of desirable attributes and achieved a state-of-the-art on the iSAID-$5^i$ and LoveDA-$2^i$ datasets. We conducted extensive ablation experiments and analyses to confirm its effectiveness and deepen our understanding of its working mechanism.

## References

[1] J. Wang, A. Ma, Y. Zhong, Z. Zheng, and L. Zhang, "Cross-sensor domain adaptation for high spatial resolution urban land-cover mapping: From airborne to spaceborne imagery," *Remote Sens. Environ.*, vol. 277, Aug. 2022, Art. no. 113058.

[2] Y. Pan, X. Wang, L. Zhang, and Y. Zhong, "E2EVAP: End-to-end vectorization of smallholder agricultural parcel boundaries from high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 246–264, Sep. 2023.

[3] Y. Wan, C. Chen, A. Ma, L. Zhang, X. Gong, and Y. Zhong, "Adaptive multistrategy particle swarm optimization for hyperspectral remote sensing image band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5520115.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.

[4] C. Zheng, J. Nie, Z. Wang, N. Song, J. Wang, and Z. Wei, "High-order semantic decoupling network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5401415.

[5] D. Chen, A. Ma, and Y. Zhong, "Semi-supervised knowledge distillation framework for global-scale urban man-made object remote sensing mapping," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103439.

[6] Z. Li, X. Zhang, and P. Xiao, "One model is enough: Toward multiclass weakly supervised remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4503513.

[7] A. Shaban, S. Bansal, Z. Liu, I. Essa, and B. Boots, "One-shot learning for semantic segmentation," in *Proc. Brit. Mach. Vis. Conf.*, Jan. 2017, pp. 167.1–167.13.

[8] N. Dong and E. Xing, "Few-shot semantic segmentation with prototype learning," in *Proc. Brit. Mach. Vis. Conf.*, Jan. 2018, pp. 1–13.

[9] M. Siam, B. Oreshkin, and M. Jagersand, "AMP: Adaptive masked proxies for few-shot segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5248–5257.

[10] J. Chen, B.-B. Gao, Z. Lu, J.-H. Xue, C. Wang, and Q. Liao, "APANet: Adaptive prototypes alignment network for few-shot semantic segmentation," *IEEE Trans. Multimedia*, vol. 25, pp. 4361–4373, 2023.

[11] K. Koffka, *Principles of Gestalt Psychology*. London, U.K.: Routledge, 2013.

[12] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[13] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*. Munich, Germany: Springer, 2015, pp. 234–241.

[14] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[15] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[16] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4096–4105.

[17] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "FarSeg++: Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 13715–13729, Nov. 2023.

[18] A. Ma, J. Wang, Y. Zhong, and Z. Zheng, "FactSeg: Foreground activation-driven small object semantic segmentation in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5606216.

[19] S. Ravi and H. Larochelle, "Optimization as a model for few-shot learning," in *Proc. Int. Conf. Learn. Represent.*, Apr. 2017, pp. 1–11.

[20] Y. Chen, Z. Liu, H. Xu, T. Darrell, and X. Wang, "Meta-baseline: Exploring simple meta-learning for few-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9062–9071.

[21] M. A. Jamal and G.-J. Qi, "Task agnostic meta-learning for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11711–11719.

[22] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[23] Q. Sun, Y. Liu, T.-S. Chua, and B. Schiele, "Meta-transfer learning for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 403–412.

[24] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," 2016, *arXiv:1606.04080*.

[25] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," in *Proc. 34th Int. Conf. Mach. Learn.*, Aug. 2017, pp. 1126–1135.

[26] E. Schönfeld, S. Ebrahimi, S. Sinha, T. Darrell, and Z. Akata, "Generalized zero- and few-shot learning via aligned variational autoencoders," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8239–8247.

[27] I. Goodfellow et al., "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.

[28] C. Lang, G. Cheng, B. Tu, and J. Han, "Learning what not to segment: A new perspective on few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8057–8067.

[29] X. Li, T. Wei, Y. P. Chen, Y.-W. Tai, and C.-K. Tang, "FSS-1000: A 1000-class dataset for few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2866–2875.

[30] W. Ren, Y. Tang, Q. Sun, C. Zhao, and Q.-L. Han, "Visual semantic segmentation based on few/zero-shot learning: An overview," *IEEE/CAA J. Autom. Sinica*, vol. 11, no. 5, pp. 1106–1126, May 2024.

[31] Z. Chang, Y. Lu, X. Ran, X. Gao, and X. Wang, "Few-shot semantic segmentation: A review on recent approaches," *Neural Comput. Appl.*, vol. 35, pp. 18251–18275, Jul. 2023.

[32] P. Tian, Z. Wu, L. Qi, L. Wang, Y. Shi, and Y. Gao, "Differentiable meta-learning model for few-shot semantic segmentation," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 12087–12094.

[33] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-shot image semantic segmentation with prototype alignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9197–9206.

[34] G. Li, V. Jampani, L. Sevilla-Lara, D. Sun, J. Kim, and J. Kim, "Adaptive prototype learning and allocation for few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8334–8343.

[35] Q. Fan, W. Pei, Y.-W. Tai, and C.-K. Tang, "Self-support few-shot semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 701–719.

[36] X. Yao, Q. Cao, X. Feng, G. Cheng, and J. Han, "Scale-aware detailed matching for few-shot aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5611711.

[37] C. Lang, J. Wang, G. Cheng, B. Tu, and J. Han, "Progressive parsing and commonality distillation for few-shot remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613610.

[38] C. Lang, G. Cheng, B. Tu, and J. Han, "Global rectification and decoupled registration for few-shot segmentation in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617211.

[39] V. Jampani, D. Sun, M.-Y. Liu, M.-H. Yang, and J. Kautz, "Superpixel sampling networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 352–368.

[40] B. Irving, "MaskSLIC: Regional superpixel generation with application to local pathology characterisation in medical images," 2016, *arXiv:1606.09518*.

[41] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[42] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.

[43] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[44] X. Zhang, Y. Wei, Y. Yang, and T. S. Huang, "SG-one: Similarity guidance network for one-shot semantic segmentation," *IEEE Trans. Cybern.*, vol. 50, no. 9, pp. 3855–3865, Sep. 2020.

[45] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," in *Proc. Neural Inf. Process. Syst. Track Datasets Benchmarks*, vol. 1. San Jose, CA, USA: Curran Associates, Inc., 2021, pp. 1–12.

[46] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[47] Z. Tian, H. Zhao, M. Shu, Z. Yang, R. Li, and J. Jia, "Prior guided feature enrichment network for few-shot segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 2, pp. 1050–1065, Feb. 2022.

[48] J. Min, D. Kang, and M. Cho, "Hypercorrelation squeeze for few-shot segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 6941–6952.

[49] B. Peng et al., "Hierarchical dense correlation distillation for few-shot segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 23641–23651.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.



**Weihao Shen** (Graduate Student Member, IEEE) received the B.S. degree from the School of Geographic Information Science, China University of Geosciences, Beijing, China, in 2022. He is currently pursuing the M.S. degree with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China.

His major research interests include remote sensing image processing and computer vision, with a focus on visual language understand and few-shot semantic segmentation.



**Ailong Ma** (Senior Member, IEEE) received the B.S. degree from China University of Petroleum, Qingdao, China, in 2010, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2017.

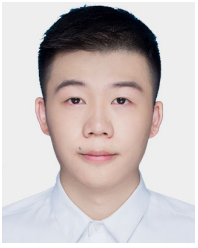
He is currently an Associate Professor with Wuhan University. He has published more than 30 research articles in *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION. His major research interests include multimodal remote sensing image processing, pattern recognition, and disaster response.



**Junjue Wang** (Student Member, IEEE) received the B.S. degree from the School of Geography and Information Engineering, China University of Geosciences, Wuhan, China, in 2019, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, in 2024.

His major research interests include multimodal remote sensing data interpretation.

Dr. Wang won the First-Place Prize in the 2022 LandSlide4Sense Contest, the Second-Place Prize in Single-View Semantic 3-D Challenge of 2019 IEEE GRSS Data Fusion Contest, and the Fourth-Place in xView2 Challenge.



**Zhuo Zheng** (Member, IEEE) received the B.S. degree from the School of Geography and Information Engineering, China University of Geosciences, Wuhan, China, in 2018, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, in 2023.

He is currently a Postdoctoral Researcher with Stanford Artificial Intelligence Laboratory (SAIL), Department of Computer Science, Stanford University, Stanford, CA, USA. He has published over ten first-author papers in leading journals and conferences, such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IJCV, RSE, ISPRS P&RS, NeurIPS, IEEE CVPR, ICCV, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His major research interests include Earth vision and simulation, especially multimodal and multitemporal remote sensing image analysis.

Dr. Zheng won the Second Place Prize in the Single-View Semantic 3-D Challenge of the 2019 IEEE GRSS Data Fusion Contest, the Fourth Place Overall in xView2 Challenge, the Top Graduate Award in SpaceNet 6 and EarthVision Workshop Challenge at CVPR 2020, the Fourth Place in the Multitemporal Semantic Change Detection Challenge of the 2021 IEEE GRSS Data Fusion Contest, and the Fifth Place and Model Write-Up Bonus Award in Overhead Geopose Challenge. He was a first-place recipient of the 2021 John I. Davidson President's Award.



**Yanfei Zhong** (Senior Member, IEEE) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He organized the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) Research Group. He has published more than 150 research papers in international journals, such as *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Dr. Zhong is a fellow of the Institution of Engineering and Technology (IET). He is currently serving as an Associate Editor for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and the *International Journal of Remote Sensing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 04:30:55 UTC from IEEE Xplore. Restrictions apply.